

Sean Luka Girgis

Data Engineer / Technical Support | Python, SQL, AWS Glue, ETL & Data Quality

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, AWS Glue/S3/Athena/Redshift, PySpark, ETL/ELT pipelines, data modeling, analytics reporting, data quality, RCA, production troubleshooting, governance-aware delivery, and stakeholder support. Strong background building and operating reliable data pipelines, integrating data from multiple monitoring and operational sources, validating accuracy and completeness, tuning SQL/reporting layers, and debugging user-reported data issues within enterprise SLA-style environments. Best aligned to data engineering roles requiring hands-on SQL/Python, ETL support, AWS data workflows, pipeline reliability, user/technical support, data modeling, stakeholder communication, RCA documentation, and production analytics enablement; Snowflake, BigQuery, PostgreSQL/MySQL, Lambda, and Step Functions are positioned as adjacent or transferable areas unless explicitly supported by prior work or active study.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Designed, built, and supported Python/Pandas/PySpark and SQL ETL workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for analytics, reporting, forecasting, and operational decision support.
- Built AWS and warehouse-backed data platform components using S3 landing patterns, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and analytics.
- Created data lake, warehouse, and data mart-style reporting structures that transformed raw operational feeds into reusable, validated datasets for technical and business stakeholders.
- Used complex SQL across Oracle, Amazon Redshift, and reporting structures to validate transformations, troubleshoot discrepancies, tune queries, and improve downstream reporting performance.
- Implemented data quality and reconciliation checks for row counts, nulls, duplicates, unmatched joins, freshness, abnormal deltas, and output comparisons against trusted prior runs.
- Debugged and analyzed data issues raised by users and stakeholders by tracing source feeds, ETL logic, SQL transformations, metric behavior, failed outputs, and downstream reporting discrepancies.
- Produced clear RCA notes, status updates, metric definitions, source mappings, runbooks, and support documentation to explain issue impact, resolution, and prevention steps.
- Partnered with infrastructure, application, analytics, engineering, and business teams to understand data needs, deliver reliable reporting solutions, and improve operational efficiency.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Analyzed real-user and synthetic monitoring datasets to identify data quality issues, performance trends, reliability risks, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services monitoring and reporting environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and documentation used by application, middleware, infrastructure, and operations teams.
- Provided technical guidance, onboarding support, training material, and knowledge sharing to improve adoption of monitoring, reporting, and operational standards.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration validation for a high-throughput shopping engine moving from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported schema conversion, data validation, phased cutover, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, KornShell/ksh, C++, C, Java concepts

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, curated transformations, analytics datasets, query optimization, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, duplicate/unmatched record checks, missing feed diagnosis, and prior-run comparisons.

Technologies: SQL, Python, Pandas, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting and Data Platform

Built Python/Pandas/PySpark workflows to process telemetry, validate outputs, prepare forecasting datasets, and produce trusted reporting for capacity and risk discussions.

Technologies: Python, Pandas, PySpark, SQL, Prophet, Data Validation

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation